

Material Test Report / Certificate TSS-25-00981

Tri-State Superalloys Corp. | 2 Forge Park Dr, Monroe, NC 28110

Certificate No. / Date	TSS-25-00981 / 2025-03-07
Heat / Batch No.	HT-71834
Customer	American Completion Tools, Inc.
Purchase Order	ACT PO-88104
Product	Forged bar, 6.25 in. dia, solution treated & double aged
Specification(s)	ASTM B637 UNS N07718; AMS 5662; API 6ACRA; NACE MR0175/ISO 15156-3
Quantity	6 bars / 4,480 lb

Chemical Composition (wt %)

Element	Value	Element	Value
Ni	52.8	Al	0.55
Cr	18.4	Co	0.22
Nb+Ta	5.12	C	0.031
Mo	3.02	B	0.004
Ti	0.98	Fe	Bal.

Mechanical / Physical Properties

Property	Result	Requirement
Yield strength (0.2% offset)	152,900 psi	>= 120,000 psi
Tensile strength	191,400 psi	>= 150,000 psi
Elongation in 4D	24 %	>= 20 %
Reduction of area	41 %	>= 35 %
Hardness	38 HRC	<= 40 HRC (MR0175)
Charpy V-notch @ -75 degF (avg of 3)	48 ft-lb	>= 30 ft-lb

Thermal Treatment

Solution 1,875 degF 1.5 hr, water quench; aged 1,325 degF 8 hr, furnace cool to 1,150 degF, hold 8 hr, air cool (per AMS 5662 / API 6ACRA)

NDE / Additional Testing

UT per ASTM A388 S1; macro-etch per ASTM E381: accepted; grain size 4-6 per ASTM E112

Sour Service (NACE MR0175 / ISO 15156) Statement

Certified NACE MR0175/ISO 15156-3 Annex A.2 nickel-based alloys, hardness 40 HRC maximum: COMPLIANT.

Remarks

Double vacuum melted (VIM + VAR). Certificate per EN 10204 Type 3.1. Used by ACT for: PKR-7000 mandrels and subs (BOM items ACT-PKR-7000-01/-02/-10) and HPHT fasteners.

Certification

We hereby certify that the material described above was manufactured, sampled, tested and inspected in accordance with the referenced specifications and was found to meet those requirements. This certificate is issued in accordance with EN 10204 Type 3.1.

Role	Name	Signature	Date
Authorized Signatory	E. Kowalski, Director of Quality, Tri-State Superalloys	/signed electronically/	2025-03-07
ACT Receiving QA Review	S. Nguyen	/signed electronically/	2025-03-07